Junyi Zhu

Email: junyizhu.ai@gmail.com | Homepage: junyizhu-ai.github.io | Google Scholar

ABOUT ME

Senior Applied Scientist at Microsoft UK with a PhD in Artificial Intelligence from KU Leuven. Experienced in training generative AI—including large language models and image generation models, inference/test-time algorithms, as well as privacy-preserving machine learning. Author of multiple papers at leading AI conferences such as NeurIPS, ICML, ICLR, CVPR, and EMNLP. Passionate about advancing AI systems from research to real-world applications, bridging cutting-edge methods with scalable, impactful solutions.

SKILLS

Machine Learning & AI

Reinforcement Learning (RL); Large Language Models (LLM); Retrieval-Augmented Generation (RAG); Diffusion Models; Text and Image Embedding Models; Federated Learning; Differential Privacy (DP)

Language

English; Chinese; German

EMPLOYMENT EXPERIENCE

MICROSOFT UK

Senior Applied Scientist

London, UK • Jan 2025 – Present

- Drive post-training efforts for Microsoft 365 Copilot, where I focus on building models that possess deep contextual awareness of workplace users, their roles, and their organizations.

SAMSUNG RESEARCH UK

Senior Researcher

London, UK • Sep 2024 – Dec 2025

- Led research and prototyping of mobile language models for personalized on-device AI systems.
- Designed and implemented scalable federated learning pipelines for privacy-preserving neural network training.

KU LEUVEN, ESAT-PSI LAB

Research Associate

Leuven, Belgium • Apr 2024 – Sep 2024

- Researched AI search systems and efficiency of training and inference in image generative models.

RESEARCH COLLABORATIONS & PROJECTS

Global/Remote • Oct 2023 – Jan 2026

- **MemTensor**:Post-training large language models with reinforcement learning and building AI search systems based on retrieval-augmented generation (RAG).
- **Microsoft Research (Redmond)**: Collaborated on improving diffusion-based image generation and developing unified principles for generative modeling, representation learning, and classification.
- **Infinigence-AI**: Worked on improving training and inference efficiency for large language models and diffusion models.

ACADEMIC ENGAGEMENT

AREA CHAIR	CVPR 2026
REVIEWER	ICML, NeurIPS, ICLR, CVPR, ACL, EMNLP, ICCV, TMLR (2021–2025)
VISITING RESEARCHER	NICS-EFC Lab, Tsinghua University (Aug–Oct 2023)

TALKS (SELECTED)

ICLR TINY PAPER TRACK (2024)

Rescaling Intermediate Features Makes Trained Consistency Models Perform Better

LEUVEN.AI JUNIOR RESEARCHERS DAY (2023)
Confidence-aware Personalized Federated Learning via Variational Expectation Maximization

NEURIPS WORKSHOP (2022)
Tackling Personalized Federated Learning with Label Concept Drift

SELECTED PUBLICATIONS

*Full list on Google Scholar

- Wang, Z.*, **Zhu, J.***, Tang, B., Li, Z., et al. *Jigsaw-R1: A Study of Rule-based Visual Reinforcement Learning with Jigsaw Puzzles*. arXiv, 2025. (*co-first*)
- Lin, Z., Liu, E., Ning, X., **Zhu, J.**, et al. *Latent Zoning Network: A Unified Principle for Generative Modeling, Representation Learning, and Classification.* NeurIPS, 2025.
- **Zhu, J.**, Ozkan, S., Maracani, A., Mutlu, S., et al. *Multi-Task Pre-Finetuning of Lightweight Transformer Encoders for Text Classification and NER*. EMNLP, 2025.
- Wang, Y., Wang, P., Xi, C., Tang, B., **Zhu, J.**, et al. *Adversarial Preference Learning for Robust LLM Alignment.* ACL Findings, 2025.
- Tang, B.*, **Zhu, J.***, Wang, H., Xi, C., et al. *Xinyu AI Search: Enhanced Relevance and Comprehensive Results with Rich Answer Presentations*. arXiv, 2025. (*co-first*)
- Liu, E.*, **Zhu, J.***, Lin, Z., Ning, X., et al. *Linear Combination of Saved Checkpoints Makes Consistency and Diffusion Models Better*. ICLR, 2025. (*co-first*)
- **Zhu, J.***, Liu, S.*, Yu, Y., Tang, B., et al. *FastMem: Fast Memorization of Prompt Improves Context Awareness of Large Language Models*. EMNLP Findings, 2024. (*co-first*)
- Liu, E.*, **Zhu, J.***, Lin, Z., Ning, X., et al. *Efficient Expert Pruning for Sparse Mixture-of-Experts Language Models: Enhancing Performance and Reducing Inference Costs*. arXiv, 2024. (*co-first*)
- **Zhu, J.**, Lin, Z., Liu, E., Ning, X., Blaschko, M. B. *Rescaling Intermediate Features Makes Trained Consistency Models Perform Better*. ICLR Tiny Papers (Notable), 2024.
- **Zhu, J.**, Yao, R., Blaschko, M. B. *Surrogate Model Extension: A Fast and Accurate Weight Update Attack on Federated Learning*. ICML, 2023.
- **Zhu, J.**, Blaschko, M. B. *Improving Differentially Private SGD via Randomly Sparsified Gradients*. TMLR, 2023.
- **Zhu. J.***, Ma, X.*, Blaschko, M. B. *Confidence-aware Personalized Federated Learning via Variational Expectation Maximization*. CVPR, 2023. (*co-first*)
- **Zhu. J.**, Blaschko, M. B. *R-GAP: Recursive Gradient Attack on Privacy*. ICLR, 2021.

EDUCATION

KU LEUVEN
Ph.D., Artificial Intelligence
Leuven, Belgium

- Dissertation: Privacy-Preserving and Distributed Machine Learning
- Funded by Flemish Government (AI Research Program) and FWO Scholarship

KARLSRUHE INSTITUTE OF TECHNOLOGY
M.Sc., Information Technology and Vehicle Engineering
Karlsruhe, Germany

- Thesis: Transferring Grid Maps to Aerial Imagery for Localization in Autonomous Driving using Generative-Adversarial Networks (GANs)

DRESDEN UNIVERSITY OF TECHNOLOGY
B.Sc. (Exchange)
Dresden, Germany

- Thesis: Design and Implementation of Automated Penetration Measuring System

BEIJING INSTITUTE OF TECHNOLOGY
B.Sc., Mechanical Engineering and Automation
Beijing, China